

Practice Problems (20 Questions)

Migration (Continent-Island)

1. In the continent-island model, if $p_c = 0.8$, $p_t = 0.3$, and $m = 0.1$, calculate Δp .
2. Using the data from Q1, calculate p_{t+1} .
3. If the island initially has $p_0 = 0$, $p_c = 1$, and $m = 0.05$, what is p_1 ? p_2 ?
4. An island has $p_t = 0.2$ and $p_c = 0.7$. What migration rate m would result in $\Delta p = 0.1$?
5. A continent has $p_c = 0.9$ for a beneficial allele. An island has $p_0 = 0.1$. After 10 generations of migration with $m = 0.1$, what is the expected island frequency?
6. True or False: In the continent-island model without drift, the island always reaches p_c exactly.
7. Explain in one sentence: Why does migration oppose local adaptation?

Founder Effect

8. A rare neutral allele has frequency $q = 0.005$ in a large population. If 10 individuals found a new population, what is the probability the allele is lost?
9. Why did colorblindness become common on Pingelap but not on neighboring islands?
10. A founder event starts with $N = 4$ individuals. What is the probability that a neutral allele with frequency $q = 0.1$ in the source population is present in the founders?
11. Explain how successive founding events (like in white-eye birds) reduce genetic diversity.

Wright-Fisher Drift

12. In the Wright-Fisher model, what is the probability that a neutral allele with initial frequency 0.3 eventually becomes fixed?
13. A population of $N = 50$ starts with heterozygosity $H_0 = 0.5$. What is expected H after 100 generations?
14. Buri's experiment used $N = 16$ flies per replicate. What fraction of heterozygosity remained after 1 generation? After 10 generations?
15. True or False: In the Wright-Fisher model, the mean allele frequency across many replicate populations changes over time.
16. Why does variance in allele frequency increase over time under drift?

17. A population of size $N = 10$ starts with two neutral alleles at equal frequency (0.5 each). What is the probability that after many generations, the population is fixed for one allele? For the other?
18. Explain in one sentence: How does population size affect the rate of heterozygosity loss?

Conservation & Integration

19. In the Similipal tiger example, explain how the "black tiger" mutation spreading to 60% of the population is a signature of genetic drift, not natural selection.
20. Compare and contrast: How do migration (gene flow) and drift have opposite effects on genetic variation between populations?

Answer Key

1. $\Delta p = m(p_c - p_t) = 0.1(0.8 - 0.3) = 0.1 \times 0.5 = 0.05$
2. $p_{t+1} = p_t + \Delta p = 0.3 + 0.05 = 0.35$
3. $p_1 = (1-0.05) \times 0 + 0.05 \times 1 = 0.05$; $p_2 = (1-0.05) \times 0.05 + 0.05 \times 1 = 0.0475 + 0.05 = 0.0975$
4. $\Delta p = m(p_c - p_t) \rightarrow 0.1 = m(0.7 - 0.2) \rightarrow 0.1 = m(0.5) \rightarrow m = 0.1/0.5 = 0.2$
5. $p_t = p_c + (p_0 - p_c)(1 - m)^t = 0.9 + (0.1 - 0.9)(0.9)^{10} = 0.9 + (-0.8)(0.3487) = 0.9 - 0.279 = 0.621$
6. True (in the deterministic model with no drift, equilibrium is exactly p_c).
7. Migration brings in alleles from the continent that may be maladapted to local island conditions, swamping local beneficial alleles.
8. $P(\text{loss}) = (1 - q)^{2N} = (0.995)^{20} \approx 0.9046$ (90.5% chance of loss).
9. By chance, the typhoon survivors (founders) included a carrier of the allele. Subsequent drift and inbreeding increased its frequency – a founder effect.
10. $P(\text{present}) = 1 - (1 - 0.1)^8 = 1 - (0.9)^8 = 1 - 0.4305 = 0.5695$ (~57%).
11. Each founding event samples a small subset of the previous population's alleles. Rare alleles are easily lost by chance, reducing total diversity stepwise.
12. Probability of fixation = initial frequency = 0.3 (or 30%).
13. $H_{100} = 0.5 \times (1 - 1/100)^{100} = 0.5 \times (0.99)^{100} \approx 0.5 \times 0.366 = 0.183$
14. Per generation fraction remaining = $1 - 1/(2N) = 1 - 1/32 = 0.96875$. After 10 generations: $0.96875^{10} \approx 0.727$ (72.7% of heterozygosity remains).
15. False. The mean frequency remains constant at p_0 .
16. Replicate populations wander randomly; over time some fix one allele, some fix the other, so variance across replicates increases.
17. Probability fixed for one allele = 0.5; fixed for the other = 0.5. (Total fixation probability = 1, split equally.)
18. Smaller populations lose heterozygosity faster because $1/(2N)$ is larger.
19. The black coat mutation is neutral (no known survival advantage or disadvantage). Its rise to 60% is due to random sampling in a tiny, inbred population – exactly the Wright-Fisher prediction that drift can change frequencies of neutral alleles. If selection were driving it, we would expect a fitness difference, which has not been demonstrated.
20. Migration reduces genetic differences between populations (homogenizes). Drift increases differences between populations by randomly fixing different alleles in each.